THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET A

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The molecular/formula unit mass of water, H_2O is (H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40): 1

(a) 16 u **(b)** 36 u **(c)** 18 u **(d)** 20 u

2. The molecular/formula unit mass of hydrogen chloride, HCl is (H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40): 1

(a) 38.5 u **(b)** 34.5 u **(c)** 73 u **(d)** 36.5 u

3. 4.0 g of $CaCO_3$ reacts with 2.92 g of HCl in a closed vessel. If 1.76 g CO_2 and 0.72 g H_2O are formed, the mass of $CaCl_2$ formed is: 1

(a) 4.44 g **(b)** 6.92 g **(c)** 5.44 g **(d)** 4.04 g

4. On decomposing 18 g of pure water, the masses of hydrogen and oxygen obtained are: 1

(a) 9 g and 9 g **(b)** 4 g and 14 g **(c)** 2 g and 16 g **(d)** 1 g and 17 g

5. The chemical formula of calcium chloride is: 1

(a) $CaCl$ **(b)** $CaCl_2$ **(c)** Ca_2Cl **(d)** Ca_2Cl_2

6. The correct name of the covalent compound CS_2 is: 1

(a) carbon sulfur two **(b)** carbon disulfide
(c) monocarbon disulfide **(d)** carbon sulfide

7. The Law of Conservation of Mass was proposed by: 1

(a) Mendeleev **(b)** Antoine Lavoisier
(c) John Dalton **(d)** Joseph Proust

8. The chemical bond formed by sharing of electrons is called: 1

(a) ionic bond **(b)** hydrogen bond
(c) covalent bond **(d)** metallic bond only

9. The chlorine in NaCl is present as: 1

(a) neutral atom **(b)** Cl^+ ion
(c) Cl^- ion **(d)** Cl_2 molecule

10. Ionic compounds conduct electricity: 1

(a) in molten state and in aqueous solution **(b)** never
(c) only as gases **(d)** in the solid state only

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State the Law of Conservation of Mass. 2

12. Why did the balloon experiment (open flask) show a loss in mass while the closed set-up did not? 2

13. 4.0 g $CaCO_3$ reacts with 2.92 g HCl to give 1.76 g CO_2, 0.72 g H_2O and 4.44 g $CaCl_2$. Verify the Law of Conservation of Mass. 2

14. If 12 g carbon combines with 32 g oxygen, how much CO_2 forms from 6 g carbon? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain the Law of Conservation of Mass with the help of the reaction between sodium sulfate and barium chloride solutions. 3

16. In a closed vessel 4.0 g $CaCO_3$ and 2.92 g HCl react to give 1.76 g CO_2, 0.72 g H_2O and some $CaCl_2$. Find the mass of $CaCl_2$ formed. 3

17. A student burns 10 g of ethanol in an open beaker and finds the mass decreased. Is the law of conservation of mass violated? Explain. 3

OR

12 g of carbon combines with 32 g of oxygen to form CO_2. How much CO_2 is formed from 2.4 g of carbon? Which law is used? 3

18. Sodium chloride has Na : Cl = 23 : 35.5 by mass. How much chlorine reacts with 46 g of sodium? How much NaCl is formed? 3

OR

A compound has 40 % sulfur and 60 % oxygen by mass. How much oxygen is present in a sample with 20 g sulfur? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) State the Law of Conservation of Mass.
(ii) Describe an experiment to verify it using sodium sulfate and barium chloride.
(iii) Why is a closed system required if gas is produced? 5

OR

Case study: Students react 4.0 g calcium carbonate with 2.92 g hydrochloric acid in a closed container and obtain 1.76 g CO_2, 0.72 g H_2O and $CaCl_2$. 5
(i) Find the mass of $CaCl_2$.
(ii) Which law is verified?
(iii) What would happen to the measured mass if the vessel were open?
(iv) Why?

20. (i) State the Law of Constant Proportions and name the scientist. 5
(ii) Water from a river and from the sea each give H : O = 1 : 8. Explain.
(iii) 36 g of water is decomposed; find the masses of hydrogen and oxygen.
(iv) Does the law hold for mixtures?

OR

Case study: Sodium chloride contains sodium and chlorine in the ratio 23 : 35.5 by mass. 5
(i) How much chlorine reacts with 46 g of sodium?
(ii) How much NaCl forms?
(iii) How much sodium is in 117 g of NaCl?
(iv) Which law helps?

END OF QUESTION PAPER — **Total: 40 Marks**